

Psychology and Neuroscience 40th Annual Graham Goddard In-House Conference



April 24-25, 2014
Life Sciences Centre
Room 4258/4263

2014 Psychology and Neuroscience In-House Conference

The 2014 Psychology and Neuroscience Graham Goddard In-House Conference

Welcome! We would like to thank you for attending the 40th Annual Graham Goddard In-House Conference. This is an excellent opportunity to hear about the ongoing research of students, faculty and colleagues. It is your participation that makes this event a success.

Thanks should also be extended to the In-House Conference Committee. The time that the following individuals put into the planning and organization of the In-House is greatly appreciated: Sophie Jacques, Max Hebert, Jill King, Suzanne King, Kiera O'Neil, Kyle Roddick, Lindsay Rubinfeld and Kurt Stover

Dr. Graham V. Goddard

Dr. Graham V. Goddard (1938-1987) was a leading researcher at Dalhousie University. Goddard invented the kindling model of epilepsy that has, since its introduction in the 1960s, become widely used research tool in studies of long-term potentiation (LTP) and epilepsy. His work was published in Nature, Science, along with other high profile journals. During his time at Dalhousie, Goddard helped establish Dalhousie as a leading research institution. Goddard also organized the annual Psychology Department In-House Conference, to help investigators in the department update others about their work and encourage academic dialogue. It is in his honour that we continue this tradition today.

Friendly Reminders

Talks will be held in LSC 4253/4268.

Coffee and tea will be available in the 4th Floor Study Centre.

Lunches will take place in the 2nd Floor Lounge.

There will be a dinner and reception to follow at the end of Friday April 25th, in the 2nd Floor Lounge.

Thursday, April 24, 2014 (Room 4263)

8:30am – 10:15am: Session 1. Chair: Shelley Adamo

8:30 T1) Introduction

Tara Perrot

8:45 T2) Mate preferences in speed-dating: Do women really prefer men with higher earning prospects?

Sean Mackinnon, D.F. Selterman, E. Chagnon and M. Furda

9:00 T3) The squeeze is on: Increased competition for faculty positions has ramifications for the recruitment of women into academia.

Shelley Adamo

9:15 T4) Breastfeeding in the NICU: An exploratory study of barriers and facilitators from the perspectives of mothers and health-care professionals.

Rebecca Tucker, Patrick McGrath, Anna Huguet, Darlene Inglis, Marsha Campbell-Yeo, Michelle LeDrew

9:30 T5) Impact of Menstrual Cycle Phase on Endocrine Effects of Partial Sleep Restriction in Healthy Women

Amanda LeRoux, Lisa Wright, Tara Perrot, Benjamin Rusak

9:45 T6) Validating the Children's Sleep Habits Questionnaire Against Polysomnography in School-Aged Children

Adria Markovich, Melissa Gendron and Penny V. Corkum

10:00 T7) Sleep disturbance and daytime sleepiness increase dementia risk in healthy individuals independent of overall health status

Roxanne Sterniczuk, Benjamin Rusak, Kenneth Rockwood

10:15am – 10:45am: COFFEE BREAK (LSC 4316)

10:45am – 12:15pm: Session 2. Chair: Tara Perrot

10:45 T8) Why my Friends Drink is Why I Drink

Sherry Stewart, N. Castellanos-Ryan, F. Vitaro and P.J. Conrod

11:00 T9) Early Life Models of Anxiety: The Role of Age, Sex and the Environment

Tara Perrot and Amanda Green

11:15 T10) Is my hangover affecting my ability to learn? Neurological evidence from electroencephalography

Ashley Howse, Marcel Peloquin, Cameron Hassall and Olav Krigolson

11:30 T11) Do Catastrophizing of Bodily Sensations and Health-Related Reassurance-Seeking Maintain Health-Related Anxiety?

Cynthia Ramasubbu, Simon Sherry, Chantal Gautreau, Dayna Sherry, Logan Nealis, Heather Hadjistavropoulos, and Sherry Stewart

11:45 T12) Investigating Possible Reciprocal Relationships between Symptoms of Depression and Pathological Gambling in Emerging Adults

Annie Chinneck, Sean McKinnon and Sherry Stewart

12:00 T13) A longitudinal exploration of advantageous and disadvantageous inequity aversion in children

Amanda Williams and Chris Moore

12:15 T14) “Give It Away, Give It Away, Give It Away Now”: Children’s Sharing Generosity & Receiving Preferences in Contexts of Anonymity

Jennifer E. Richards & Chris Moore

12:15pm – 1:30pm: LUNCH AND POSTER SESSION (2nd Floor Lounge)

P1) fERN differences within Subclinical Psychopaths

Mitchell Coupland & Olav Krigolson

P2) Contrast adaptation in mouse V1 depends on spatial frequency but not firing rate

Matthew P. Lowe, Jillian L. King, Nathan A. Crowder

P3) Retinorecipient areas in the diurnal rodent *Octodon Degu*

Malik Abouleish, Kevin J. Frank, Jillian L. King, Nathan A. Crowder

P4) Investigating a Role for Matrix Metalloproteinases in the Cortex of Visually Deprived Cats.

Josh Bowdridge and Dr. Kevin Duffy

P5) Recovery from monocular deprivation in kittens induced by ten days of darkness depends upon the acuity in the non-deprived eye and requires a period of binocular vision

Matthew J. Smithen, Kaitlyn Holman, Donald Mitchell & Kevin Duffy

P6) ERP Components Underlying Prism Adaptation (PA)

Stephane Maclean, Yoko Ishigami-Doyle, Cameron Hassall, Olave Krigolson & Gail Eskes

1:30pm–3:15pm: Session 3. Chair: Patricia McMullen

1:30 T15) Assessing the Relation Between Subjective Cognitive Impairment, Neurological Functioning, and Psychosocial Symptoms in Multiple Sclerosis Patients

Max Hebert and John Fisk

1:45 T16) Face Processing and Gender Recognition: An Eye-Tracking Study

Stephanie Sobey, Dr. Patricia McMullen, Dr. Heath Matheson

2:00 T17) A Comprehensive Review of the Neural Correlates, the Mechanisms, and the Involvement of the Primary Motor Cortex (M1) in Motor Imagery

Kerry McInnes and Shaun Boe

2:15 T18) Motor imagery as a modality of skill acquisition: Assessing the neural correlates of motor imagery

S. Kraeutner, A. Gionfriddo, T. Bardouille and Shaun Boe

2:30 T19) Effects of alerting tones on prism adaptation in healthy older adults

Yoko Ishigami, Franziska Kintzel, & Gail Eskes

2:45 T20) Beyond the Physical: Can We Improve Thinking and Memory in Parkinson's?

Stephanie A. H. Jones, Janet Green and Gail Eskes

3:00 T21) Compromised Attention After Stroke? Maybe Not!

Ryan Wilson & Gail Eskes

3:15pm-3:45pm: COFFEE BREAK (Room 4316)

3:45pm-5:30pm: Session 4. Chair: Dennis Phillips

3:45 T22) Applications of scent processing with canines and other animals 1: Implementation and deployment challenges.

Simon Gadbois & Catherine Reeve

4:00 T23) Applications of scent processing with canines and other animals 2: Methodological challenges of canine olfactory psychophysics

Catherine Reeve & Simon Gadbois

4:15 T24) Song Learning Under Stress: Does single parenting affect song development in zebra finches?

Jordan Fisk, Sean Aitken, Tara Perrot, Leslie Phillmore

4:30 T25) Human Sensitivity to Single Temporal Irregularities

Laura Ovens, Dennis P. Phillips

4:45 T26) In contrast to the effects observed in kittens, 10 days of darkness does not restore neural or visual plasticity in adult cats.

Kaitlyn Holman, Kevin Duffy and Donald Mitchell

5:00 T27) Orientation Plasticity in Mouse Primary Visual Cortex

Jill King

5:15 T28) Candidate neural substrate of a motion detection circuit in the visual system of *Drosophila*

Kazunori Shinomiya and Ian A. Meinertzhagen

Friday, April 26, 2013 (Room 4263)

8:45am – 10:15am: Session 5. Chair: John Barresi

8:45 T29) Prenatal paternal stress and postnatal enhanced home cage affect maternal care and anxiety like behaviour in juvenile rats

Austin Korgan, Tara Perrot, Ian Weaver

9:00 T30) How do student-generated questions relate to comprehension, self-efficacy and achievement in an undergraduate chemistry course?

Bradley Bergey

9:15 T31) Dyadic conflict, drinking to cope, and alcohol-related problems: A 4-week longitudinal study

Laura Lambe, Sean MacKinnon, L. Cartwright, M. Korecky, M. Orr, S.J. Park, D. Song, and Sherry Stewart

9:30 T32) Alzheimer's Disease and 5XFAD Mice: A Study of Hormonal Abnormalities and Metabolism

William Gendron, Dr. Richard E. Brown, Dr. Younes Anini

9:45 T33) The moral bases of John Woolman's opposition to slavery: A comparison of qualitative and quantitative analyses.

John Barresi

10:00 T34) You Can Take a Peek Under the Covers Without Taking a p.

John Christie

10:15am – 10:45am: COFFEE BREAK (Room 4316)

10:45am – 12:15pm: Session 6. Chair: Aaron Newman

10:45 T35) Second-language vocabulary training strategies and their impact on learning outcomes and brain function

Kiera O'Neil, Max Hauser, Vivian Eng, Cheryl Frenck-Mestre and Aaron J Newman

11:00 T36) Investigating the factors that influence lexical stress processing in English disyllabic words.

Rebecca MacDonald & Erin Sparks

11:15 T37) Understanding Cortical Activation Associated with Second Language Acquisition

Thérèse M. Chevalier, Aaron J. Newman

11:30 T38) The relationship between implicit sequence learning, speech in noise perception, and working memory.

Alexandria Muise-Hennessey, Sean McWhinney, and Aaron J. Newman.

11:45 T39) Morphological and Syntactic Awareness: Understanding the Deficits in Children with Poor reading Comprehension

S. Helene Deacon, Shelley Tong, & Kate Cain

12:00 T40) Champ vs. Chump: Knowledge of Opponent Ability Modulates Reward Processing in Medial-Frontal Cortex

Ralph Redden, Cameron Hassall & Olav Krigolson

12:15 T41) Learning what matters most

Cameron Hassall and Olav Krigolson

12:30pm – 1:30pm: LUNCH AND POSTER SESSION (2nd floor lounge)

P7) Relationship of the components of the Attention Network Test-Interaction (ANT-I) to standardized measures of attention and executive function

Yoko Ishigami, Stewart Longman, Amanda Tyndall, Grazyna Burek, Melanie Denheyer, Marc Poulin, Gail Eskes

P8) Neuroeconomics: Changes in neural activity during learning

Luke Burke, Laura Alsip, Nicole Hodder, Sarah Vokey, Cameron Hassall, and Olav Krigolson

P9) A step towards building better video "games" for rehabilitation: A survey investigating entertainment preferences in adults

Joshua P. Salmon, Graham C. Wilson, Raymond M. Klein, & Gail A. Eskes

P10) Drosophila Melanogaster

Nedjma Ababou and Meagan Wiederman

P11) The Impact of Socioeconomic Status on Reward Learning in the Medial-Frontal Cortex

Ashley Howse, Marcel Peloquin, S. Doherty, Cam Hassall & Olav Krigolson

P12) Personality Differences Among Smokers and Non-Smokers

Chad Simmons and Meredith Ivany

1:30pm–3:15pm: Session 7. Chair: Gail Eskes

1:30 T42) The Effect of Visual Noise on the Magnitude of the Directed Forgetting Effect

Noha Mohamed

2014 Psychology and Neuroscience In-House Conference

- 1:45 T43) How good is your reaction time data? Lessons learned from the Dalhousie Computerized Attention Battery (DalCAB)**
Joshua P. Salmon, Stephanie A. H. Jones, Chris Wright, Michael Lawrence, Beverly Butler, Ray Klein, Gail Eskes
- 2:00 T44) Covert measurement of covert orienting: An electroencephalographic and pupillometric investigation**
Mike Lawrence
- 2:15 T45) Prosodic Factors in Directed Forgetting of Speech Errors**
Sarah Dolan, Tracy Taylor, Jonathan Fawcett
- 2:30 T46) The directed forgetting effect: The processes associated with intentionally forgetting and remembering.**
Kate Thompson, Tracy Taylor-Helmick
- 2:45 T47) The Dalhousie Computerized Attention Battery: Validation against traditional neuropsychological testing instruments**
Lindsay Rubinfeld
- 3:00 T48) Effects of child temperament and parenting on toddlers' executive functioning**
Loriann Williams, Isabel Smith and Nancy Garon

3:15pm-3:45pm: COFFEE BREAK (Room 4316)

3:45pm-5:15pm: Session 8. Chair: Penny Corkum

- 3:45 T49) Laterality of Motor Imagery-Related Brain Activity is Modulated by Real-Time Neurofeedback**
Tim Bardouille
- 4:15 T50) Integrating methods from neuroscience, experimental psychology, and computer science in the study of cognitive science**
Jason Satel
- 4:45 T51) Spelling Matters Too! How spelling practice can benefit reading skill in children**
Nicole Conrad

5:15 Closing Comments

5:30 Dinner /Reception in the 2nd Floor Lounge

April 24, 2014

T1) Introduction

Tara Perrot

A historical overview of the people and events in our department from at least 1970. This year I will focus on the contributions of Jeannette McGlone and Heather Schellinck, who will retire this year.

T2) Mate preferences in speed-dating: Do women really prefer men with higher earning prospects?

Sean Mackinnon, D.F. Selterman, E. Chagnon and M. Furda

As part of a large-scale, collaborative research project to test the reproducibility of scientific results in psychology, we sought to replicate key findings from Eastwick and Finkel (2008). Specifically, we hypothesized – contrary to evolutionary theory – that women would not display a preference for mates with higher perceived earning prospects in a live-attraction (speed-dating) paradigm, compared to men. Participants ($N=307$) went on brief, 4-minute “dates” with potential romantic partners. Participants rated dates on attraction and relationship initiation variables after speed-dating. We also followed participants for 30 days following the speed-dating to explore actual dating experiences. Results showed the perception of greater earning prospects was associated with greater romantic interest in potential partners, $r=.17$ [.05, .29], but this effect was not moderated by gender $\chi^2(1)=0.89$, $p=.46$. We successfully replicated the findings from Eastwick and Finkel (2008), suggesting both men *and* women prefer partners with higher earning prospects, contrary to evolutionary theory.

T3) The squeeze is on: Increased competition for faculty positions has ramifications for the recruitment of women into academia.

Shelley A. Adamo

The number of women faculty in the sciences has increased steadily over the last 40 years. However, relatively fewer women than men make the transition from doctoral student to faculty member. The reason for the loss of these highly skilled individuals from academic science remains unclear. Both motherhood and a bias against women in science have been hypothesized to be the main culprits. I present evidence that excessive competition for faculty positions disproportionately affects women with children. The ratio of academic positions available, given the number of doctoral students trained, has been falling over the last decade in Canada. Studies in the US have shown that women doctoral students and post docs with children are less likely to obtain a faculty position than single women or men. A comparison with medicine demonstrates how motherhood need not be a barrier to women's participation in academic science.

T4) Breastfeeding in the NICU: An exploratory study of barriers and facilitators from the perspectives of mothers and health-care professionals.

Rebecca Tucker, Patrick McGrath, Anna Huguet, Darlene Inglis, Marsha Campbell-Yeo, Michelle LeDrew

Breastfeeding is known to have health benefits for both infants and their mothers. Based on the evidence of these benefits, the World Health Organization's recommendation is to exclusively breastfeed infants for their first six months. Despite this, we continue to see low numbers of Canadian mothers that are still exclusively breastfeeding their infants at six months of age. For infants who have been admitted to the Neonatal Intensive Care Unit (NICU), the number of mothers who are still exclusively breastfeeding at six months appears to be even lower. In this study, we will be talking to both mothers and health-care professionals in an attempt to understand their experiences and get their perceptions of the challenges to, and facilitators of, breastfeeding infants who have been admitted to the NICU. We are also interested in what they suggest as possible solutions and supports that would have helped them (or their patients, in the case of health-care professionals). Ideally, this research will provide important information that can then be used to develop an interactive web-based instructional program aimed at supporting sustained breastfeeding for mothers who have had infants admitted

T5) Impact of Menstrual Cycle Phase on Endocrine Effects of Partial Sleep Restriction in Healthy Women

Amanda LeRoux, Lisa Wright, Tara Perrot, Benjamin Rusak

There is extensive evidence that sleep restriction alters endocrine function in healthy young men, increasing afternoon cortisol levels and modifying levels of other hormones that regulate metabolism. Recent studies have confirmed these effects in young women, but have not investigated whether menstrual cycle phase influences these responses. The effects on cortisol levels of limiting sleep to 3 h for one night were assessed in two groups of women at different points in their menstrual cycles: mid-follicular and mid-luteal. Eighteen healthy, young women, not taking oral contraceptives (age: 21.8 ± 0.54 ; BMI: 22.6 ± 0.63 [mean $\pm$ SEM]), were studied. Baseline sleep durations, eating habits and menstrual cycles were monitored. Salivary samples were collected at six times of day (08:00, 08:30, 11:00, 14:00, 17:00, 20:00) during two consecutive days: first after a 10 h overnight sleep opportunity (Baseline) and then after a night with a 3 h sleep opportunity (Sleep Restricted). All were awakened at the same time of day. Women in the follicular phase showed a significant decrease ($p = 0.004$) in their cortisol awakening responses (CAR) after sleep restriction and a sustained elevation in afternoon/evening cortisol levels ($p = 0.005$), as has been reported for men. Women in the luteal phase showed neither a depressed CAR, nor an increase in afternoon/evening cortisol levels. Menstrual cycle phase dramatically altered the responses of healthy, young women to a single night of sleep restriction, implicating effects of spontaneous changes in endocrine status on metabolic responses to sleep loss.

T6) Validating the Children's Sleep Habits Questionnaire Against Polysomnography in School-Aged Children

Adria Markovich, Melissa Gendron and Penny V. Corkum

Sleep is a vital physiological behaviour in children's development, and as such it is important to be able to efficiently and accurately assess whether children display difficulties with sleep quality and quantity. The Children's Sleep Habits Questionnaire (CSHQ; Owens, Spirito, & McGuinn, 2000) is one of the most commonly used diagnostic tools for pediatric sleep problems. However, this instrument has never been validated directly against the gold standard of sleep measurement (i.e., polysomnography; PSG), and studies comparing it to actigraphy are limited. The current study assessed the validity of the CSHQ via direct comparison with PSG for 30 typically developing school-aged children (ages 6 to 12). Significant correlations were expected between the four CSHQ subscale scores that were analyzed (i.e., Sleep Onset Delay, Sleep Duration, Night Wakings, and Sleep Disordered Breathing) and related sleep parameters as measured via PSG. Furthermore, these CSHQ subscales were predicted to accurately identify 80% of children meeting related PSG criteria for a sleep disorder. Surprisingly, Pearson's correlation analyses revealed no significant correlations between CSHQ subscales and PSG variables. In addition, sensitivity and specificity analyses of the CSHQ subscales revealed consistently low sensitivity and high specificity. Combined, the CSHQ Sleep Onset Delay, Sleep Duration, Night Wakings, and Sleep Disordered Breathing subscales showed low construct validity and diagnostic validity. These results demonstrate weak support for the overall clinical utility of the CSHQ as a comprehensive screening tool for sleep difficulties in school-aged children.

T7) Sleep disturbance and daytime sleepiness increase dementia risk in healthy individuals independent of overall health status

Roxanne Sterniczuk, Benjamin Rusak, Kenneth Rockwood

This study examined the relationship between sleep disturbance and overall health status, in the development of incident dementia. Secondary analyses were conducted on cognitively healthy older males who participated in the Honolulu-Asia Aging Study (N = 3,852). A 9-item 'sleep disturbance index' consisting of factors related to nighttime insomnia and daytime sleepiness, was compared to a 60-item 'frailty index' (FI) consisting of various health-related factors. Controlling for age, education, body mass index, and FI, the 'sleep disturbance index' was significantly predictive of incident dementia after an average of 2.9 years. Individual items 'waking up several times at night' and 'sleepy most of the day', were strong predictors in risk models that included the FI; however the FI was not a significant predictor in our models. Sleep disturbance appears to be an important risk factor in the preclinical stage of incident dementia (i.e., prior to the manifestation of cognitive decline).

T8) Why my Friends Drink is Why I Drink

Sherry Stewart, N. Castellanos-Ryan, F. Vitaro and P.J. Conrod

Peers may influence one another's drinking behavior through social influences on drinking motives. A sample of 2,659 adolescents (median age = 14.5 at baseline) in 20 schools across 11 boroughs of London, UK participated. At each of 4 waves, spaced at 6 month intervals, participants completed measures of their own risky internal drinking motives and their own binge drinking frequency. At each wave, students also nominated up to five students who were their closest friends. We created two scores for each individual representing coping motives and enhancement motives in the peer group by averaging the coping motives and enhancement motives scores of his/her nominated friends. Using cross-lagged analyses we showed that, over time, the internal drinking motives of the peer group influenced individual binge drinking by way of influencing individual internal drinking motives. Moreover, peer influences on drinking motives occurred later in adolescence for coping motives than for enhancement motives.

T9) Early Life Models of Anxiety: The Role of Age, Sex and the Environment

Tara Perrot and Amanda Green

The early environment (prenatal and postnatal) shapes stress response systems, ultimately affecting adult brain and behaviour. We have developed a novel model of prenatal stress coupled with postnatal enrichment to investigate the expression of anxiety-like behaviours in adult rats. Moms were exposed to a live cat 3 times a day for each day (prenatal stress; or a control condition) during the last week of gestation and then either housed in standard cages or enhanced home cages. Offspring behaviour was examined in the juvenile period as well as in adulthood. Anxiety was reduced by prenatal stress in juvenile males, but not females, but effects were gone by adulthood. Prenatal stress also reduced play behaviour in the juvenile period. The combination of prenatal stress and postnatal enrichment produced complex results.

T10) Is my hangover affecting my ability to learn? Neurological evidence from electroencephalography

Ashley Howse, Marcel Peloquin, Cameron Hassall and Olav Krigolson

While there is strong evidence for a reward processing system involved with learning within the human medial-frontal cortex, a lot remains unclear about the factors that influence the functional efficacy of this system. Behavioural evidence suggests that alcohol hangover impacts motor control and cognitive functioning (Cherpitel et al., 1998; Verster, 2007). In the present study, we aimed to provide the first neurological evidence of an impairment of the reward processing system of the human medial frontal cortex during alcohol hangover. Participants completed a learnable gambling task while EEG data were recorded. In line with previous work, we found that the feedback error-related negativity (fERN; Miltner et al., 1997) was elicited by wins and losses during this paradigm. Further, and importantly, a reduction in fERN amplitude was observed between hangover and control participants. Additionally, a moderate correlation was observed between fERN magnitude and hangover severity. These data indicate the magnitude of the impairment increases with hangover severity. The results of the present study are important for public education and safety, as an impairment of the reward processing system may implicate individuals' ability to execute corrective behaviours—processes necessary for successfully executing potentially dangerous tasks such as driving a motor vehicle.

T11) Do Catastrophizing of Bodily Sensations and Health-Related Reassurance-Seeking Maintain Health-Related Anxiety?

Cynthia Ramasubbu, Simon Sherry, Chantal Gautreau, Dayna Sherry, Logan Nealis, Heather Hadjistavropoulos, and Sherry Stewart

The impact of catastrophizing of bodily sensations (i.e., rumination about, overconcern with, and intolerance of bodily sensations) and health-related reassurance-seeking (i.e., seeking care from others regarding one's perceived health problems) on health-related anxiety (i.e., anxiety about perceived health problems) is unknown. We hypothesized catastrophizing of bodily sensations and health-related reassurance-seeking maintain health-related anxiety, and catastrophizing of bodily sensations predicts health-related reassurance-seeking. A sample of 155 young adults was studied using a 28-day daily diary study with longitudinal follow-up. We found results consistent with two hypotheses: Catastrophizing of bodily sensations maintained health-related anxiety, and catastrophizing of bodily sensations predicted health-related reassurance-seeking. A cyclical, self-perpetuating pattern was observed wherein catastrophizing of bodily sensations and health-related anxiety contribute to one another over time.

T12) Investigating Possible Reciprocal Relationships between Symptoms of Depression and Pathological Gambling in Emerging Adults

Annie Chinneck, Sean McKinnon and Sherry Stewart

Based on literature linking emotional disorders and problem gambling, the present study tested bidirectional relationships between depression and pathological gambling in emerging adults. It was hypothesized that depressive and pathological gambling symptomology would be positively related at baseline, and that depression would confer risk for increased gambling over time, and vice versa. A sample of 517 Manitobans aged 18 to 20 years completed a series of questionnaires over four waves and five years (2007 to 2011) as part of the Manitoba Longitudinal Study of Young Adults. Data were analysed using inflated zero univariate growth curves. Depression and pathological gambling were positively related at Wave 1. Furthermore, having symptoms of depression at Wave 1 was related to a significant increase in pathological gambling symptoms over time. The reverse was not supported, however, and engaging in pathological gambling at Wave 1 was not related to changes in depression symptomology over time. Depression may lead to escalations in gambling problems over time via self-medication. This has treatment implications in that, while depression affects changes in problem gambling, the reverse is not true. Young adults who are depressed may therefore be a subgroup who should be targeted for problem gambling prevention.

T13) A longitudinal exploration of advantageous and disadvantageous inequity aversion in children

Amanda Williams and Chris Moore

How children make decisions in situations of disadvantageous inequity, (DI) where one can split resources equally or allow one's partner to receive more, has received little attention. Though research has explored decisions made in situations of advantageous inequity, (AI) -such as sharing- more thoroughly, little is known regarding how the developmental trajectories of these two forms of inequity aversion compare, and whether they are related to one another. In this longitudinal study, two forms of AI, and two forms of DI (cost and no cost) were explored in children at three time points; 4.5 years, 5 years, and 5.5 years of age. Using a forced choice resource allocation task, relationships between the two forms of each type of inequity aversion were investigated, and relationships across the three time points for AI, and DI were examined independently. Finally, the relationship between AI and DI was explored. For both AI and DI, cost and no cost variables were correlated at each time point. Correlations across all time points were observed in AI trials, and grew stronger with age, while only one relationship was observed over time in DI trials (between times one and two). A positive relationship (indicating a preference for equity) was observed between AI and DI at time one, whereas no relationships were observed later in development. These results suggest that AI and DI do indeed have distinct developmental trajectories, and that the relationship between them changes with age

T14) “Give It Away, Give It Away, Give It Away Now”: Children’s Sharing Generosity & Receiving Preferences in Contexts of Anonymity

Jennifer E. Richards & Chris Moore

People engage in prosocial behaviours, such as sharing, from a very young age. Numerous factors, including anonymity, can influence sharing. While anonymity has been shown to decrease adults’ generosity, it remains unknown whether anonymity influences children’s generosity. Two experiments examined the impact of anonymity on children’s sharing behaviour. Experiment 1 investigated if children understood anonymity, and whether their generosity was influenced when they shared as anonymous or identified donors. Experiment 2 examined whether children preferred to receive from donors who wished to be anonymous, or from donors who wished to be identified to the child. Findings from these experiments suggest that children understood the concept of anonymous sharing, and preferred to receive from identified donors. However, unlike adults, children’s generosity did not decrease when sharing anonymously.

T15) Assessing the Relation Between Subjective Cognitive Impairment, Neurological Functioning, and Psychosocial Symptoms in Multiple Sclerosis Patients

Max Hebert and John Fisk

Multiple Sclerosis (MS) is a degenerative neurological disorder that impacts quality of life. Among the symptoms, 65% of patients experience cognitive impairments (e.g., working memory, processing speed) throughout the course of their illness. Complicating this, many MS patients experience subjective mood symptoms of depression and anxiety that can distort accurate knowledge of cognitive impairments, resulting in patients concluding that their cognitive impairments are more severe than has been objectively measured. The current study will investigate subjective and objective cognitive impairments and how subjective mood can moderate this relationship. Participants of the study will be 354 adults with clinically definite MS from the Dalhousie MS Research Unit database. Moderation analysis will determine if anxiety or depression ratings act as a moderator between subjective and objective cognitive performance. Further analysis will employ structural equation modeling to determine whether subjective mood or other assessment measures of disability will be effective in accounting for the variability between subjective and objective cognitive impairment ratings.

T16) Face Processing and Gender Recognition: An Eye-Tracking Study

Stephanie Sobey, Dr. Patricia McMullen, Dr. Heath Matheson

Purpose: During social interactions, the eyes provide information that can be used to facilitate effective communication, and even play a role in identifying gender. While some research has demonstrated that mutual eye contact facilitates gender processing, others report conflicting results. It is also widely accepted that inverted faces are harder to identify, and this Face Inversion Effect is caused by a lack of expertise. This lack of expertise causes a shift to featural-based processing as opposed to the holistic processing applied to upright faces. The current study incorporates eye-tracking technology to further investigate orientation and gaze direction effects on gender processing.

Methods: Participants completed a gender categorization task in which face orientation (upright/inverted) and gaze direction (direct/averted) were manipulated. An eye-tracking device recorded saccades and fixations within specified interest areas as well as reaction time and decision accuracy. It is well established that humans are highly efficient at identifying gender; therefore, faces were morphed into varying degrees of masculinity/femininity to make the task more difficult and ensure rich eye-tracking data.

Preliminary Results: Upright faces were indeed processed faster and more accurately than inverted faces and eye-tracking showed a strong preference for the upper portion of the face. Gaze direction influenced reaction times such that averted gaze was processed faster than direct gaze – an effect that was modulated by the degree of masculinity/femininity of the face. These results suggest that mutual eye contact may interfere with gender processing and this interference is most pronounced in highly masculine faces.

T17) A Comprehensive Review of the Neural Correlates, the Mechanisms, and the Involvement of the Primary Motor Cortex (M1) in Motor Imagery

Kerry McInnes and Shaun Boe

Motor Imagery (MI) is the mental rehearsal of a movement without the actual overt execution. MI and motor execution (ME) activate overlapping brain regions. Therapeutic interventions targeting lost motor function from neural damage capitalize on ME. In other words, ME drives activation of central motor pathways, which rewires the damaged networks and rehabilitates lost motor function. Given the overlapping neural activation during ME and MI, it follows that MI also has the potential to rehabilitate compromised motor function. The following review explores the similarities and differences between MI and ME. It focuses on the theories for understanding MI —specifically, its neural correlates and mechanisms. Our understanding of MI, however, is greatly hindered by the debated findings regarding the involvement of the primary motor cortex (M1) during MI. This review therefore explores MI theories with an emphasis on the contribution of M1 during MI.

T18) Motor imagery as a modality of skill acquisition: Assessing the neural correlates of motor imagery

Sarah Kraeutner, A. Gionfriddo, T. Bardouille and Shaun Boe

Motor imagery (MI) is a form of practice in which an individual mentally rehearses a motor task, driving plastic changes in the brain and facilitating skill acquisition in the absence of physical practice. Previous research suggests that MI recruits similar brain regions as motor execution (ME), however, this data remains controversial due to methodological limitations. The current research seeks to further establish MI as a secondary modality of skill acquisition by providing electrophysiological evidence, using MEG, of an overlap between brain areas recruited for ME and MI. Participants performed both execution and imagery of a button press task. We demonstrate that regions recruited during ME were also recruited during MI. In contrast with ME, however, spatial extent of activation was greater and additional areas were recruited during MI, suggesting that MI may be a learned skill in itself. These findings further establishes MI as a secondary form of skill acquisition.

T19) Effects of alerting tones on prism adaptation in healthy older adults

Yoko Ishigami, Franziska Kintzel & Gail Eskes

One therapeutic approach for patients with spatial neglect is prism adaptation (PA), which targets the orienting deficit in spatial neglect. In the current study, we examined whether combining the PA technique with alerting tones (to enhance vigilance) increases the PA effects in older adults. Forty healthy older participants were randomly assigned to one of four conditions that varied in the presence of PA and frequency of alerting tones during the adaptation period. During the adaptation period, participants pointed towards targets while wearing clear or prismatic goggles shifting the visual field rightwards with alerting tones occurring on 0 – 50% of trials. Outcome measures included subjective-midline tasks (SMTs) and visual search. The greatest after-effects were observed with the highest frequency of the tones (50%) in an SMT involving both visual and proprioceptive guidance. We conclude that alerting tones may improve some aspects of PA treatment in patients with spatial neglect.

T20) Beyond the Physical: Can We Improve Thinking and Memory in Parkinson's?
Stephanie A. H. Jones, Janet Green and Gail Eskes

Individuals with Parkinson's Disease (PD) experience problems with working memory (WM), even in the early stages of the disease. However, no standard treatment for WM deficit is available. We sought to investigate the feasibility and preliminary effectiveness of an intensive and focused computerized WM training program for individuals in the early stages of PD. A controlled single blind randomized crossover design was used. One group receives a 'dual n-back' working memory training task first, and the second group receives a control visuospatial computer task first. Training includes at home 25 minute sessions, 5 days/week for 5 weeks. After 5 weeks, the training conditions are crossed over for phase 2. Outcome measures include WM, fluid intelligence and self-report functional outcomes immediately after each phase, and after 3 months. Preliminary pre and post training results on selected outcome measures obtained from 3 participants (2 additional participants are currently enrolled) will be presented.

T21) Compromised Attention After Stroke? Maybe Not!
Ryan Wilson & Gail Eskes

Deficits in attention control, an aspect of working memory (WM), are considered one of the most prominent impairments caused by stroke, but few studies have examined which specific aspects of attention control are affected. This study tested the ability of 15 stroke survivors and 20 healthy control subjects to ignore distracting information while completing the n-back WM task under increasing levels of load. We hypothesized increased distractor processing, i.e., longer reaction times (RT) and higher error rates (ER), for stroke survivors and healthy controls as WM load increased, and that stroke survivors would be more adversely affected by increased WM load than the healthy controls. Contrary to our hypotheses stroke survivors and healthy controls evidenced equivalent distractor processing, suggesting perseveration of some aspects of attention control after stroke. Furthermore, both groups showed *reduced* distractor processing during the most difficult task compared to all other tasks. Implications will be discussed.

T22) Applications of scent processing with canines and other animals 1: Implementation and deployment challenges.
Simon Gadbois & Catherine Reeve

Training dogs and other animals (e.g., bees, wasps) for scent detection, discrimination, identification, trailing, and tracking is an emerging field of research with its own challenges. Our lab has explored this field of research for eight years and worked with private and public institutions on specific problems (mostly environmental and medical). We will present the implementation challenges of this research, and the conflicts between the academic and R&D approach. We will then discuss future opportunities and applications and explain why we are expanding our research program to other scenting species.

T23) Applications of scent processing with canines and other animals 2: Methodological challenges of canine olfactory psychophysics

Catherine Reeve & Simon Gadbois

After eight years of trial and error, fine-tuning of procedures and training protocols, our lab has recently focused on a few techniques. We are challenging some common approaches based on psychophysical (stimulus saliency), cognitive (working memory), and neuroscience (olfactomotor activation) principles. Data will be presented to support our new approaches especially in relation to the detection of hypoglycaemia in breath samples from individuals with Type 1 diabetes.

T24) Song Learning Under Stress: Does single parenting affect song development in zebra finches?

Jordan Fisk, Sean Aitken, Tara Perrot, Leslie Phillmore

Young male songbirds such as zebra finches learn their song from listening to and memorizing their father's song and then practicing until they produce a "good copy". Memorization begins at about 25 days post-hatch, before the birds can produce song, active practice begins at about 35 days, and song crystallizes at adulthood - about 90 days. We were interested in whether having a single father throughout development, when typically both the mother and the father contribute to offspring care, would affect a son's ability to produce a good copy of adult song. To do this we removed the father or mother four days after eggs hatched, allowed birds to mature, and then compared recordings of adult offspring to their father. We found that birds reared without a model (single mother) had poor copies of fathers song as expected, but birds reared with a single father were not as good at copying father's song on several measures compared to birds reared with both parents.

T25) Human Sensitivity to Single Temporal Irregularities

Laura Ovens, Dennis P. Phillips

There is remarkably little evidence on human sensitivity to auditory temporal irregularities. This is important because deficits in neural timing are implicated in poor perception in auditory pathologies. Building on previous work using the jitter paradigm, experiments will measure human sensitivity to single temporal irregularities within 25-click trains having inter-click intervals (ICI) between 5 and 160 ms. Thresholds will be measured as a function of ICI in four different conditions which vary across two dimensions; direction of click shift (backward or forward in time), and position of subsequent clicks (block shifted or not). Block shifting of subsequent clicks results in a stimulus with only one aberrant ICI. Stimuli will be presented in two forms; as wide-band clicks, as well as high-pass clicks in the presence of a low frequency noise masker. It is expected that performance will vary as a function of ICI, as has been seen in previous work. It is further expected that thresholds will vary between conditions, with higher thresholds on conditions containing only one aberrant ICI.

T26) In contrast to the effects observed in kittens, 10 days of darkness does not restore neural or visual plasticity in adult cats.

Kaitlyn Holman, Kevin Duffy and Donald Mitchell

Recent investigations show that 10days of darkness imposed on kittens can rejuvenate cortical plasticity and promote fast recovery of vision in kittens with stable, experimentally induced amblyopia. Here, we report effects of darkness on neural and visual plasticity in adult cats. First, we report the effect of darkness on the vision of adult cats with stable, experimentally induced amblyopia. Second, we report the effect of darkness on the vision of a normal adult cat when imposed before a period of monocular deprivation (MD). Third, we report effects of darkness on recovery of soma size and neurofilament immunolabeling in the lateral geniculate nucleus (LGN) of adult cats that had experienced long-term MD. Finally, we report effects of darkness on LGN soma size and neurofilament immunolabeling in normal adult cats when imposed before MD. In all cases, 10days of darkness did not promote restoration of neural or visual plasticity in adult cats.

T27) Orientation Plasticity in Mouse Primary Visual Cortex

Jill King

Information processing in the visual system is shaped by recent stimulus history, such that prolonged viewing of an adaptor can alter the perception of subsequently presented stimuli and modify the visual response properties of neurons in multiple brain areas. In the tilt-aftereffect, the perceived orientation of a grating is often repelled away from the orientation of a previously viewed adaptor grating. A potential neural correlate for this tilt-aftereffect has been described in cat and macaque primary visual cortex (V1), where adaptation produces repulsive shifts in the orientation tuning curves of V1 neurons. In these animals, columns of V1 neurons are clustered into iso-orientation domains and pinwheel centers, and there is evidence that orientation adaptation is more prevalent near pinwheel centers. We investigated orientation adaptation in mouse V1 with the aim of establishing a genetically tractable model to study this form of plasticity. We were also curious about how orientation adaptation might differ in a species known to lack pinwheel columnar organization and have a lower proportion of orientation tuned neurons. We used stimulus protocols that were readily comparable with previous studies, and found that orientation adaptation caused repulsive shifts in the majority of orientation tuned neurons, and a general decrease in firing in neurons that were not selective for orientation.

T28) Candidate neural substrate of a motion detection circuit in the visual system of *Drosophila*

Kazunori Shinomiya and Ian A. Meinertzhagen

In the fruit fly's visual motion pathways, two cell types - T4 and T5 - are the first known relay neurons to signal small-field direction-selective motion responses. I report that multiple types of transmedulla (Tm) neurons provide inputs to T5 at their terminals in the third neuropil, or lobula. From L2 cells, the OFF-channel neuron in the lamina, motion information comes from three types of Tm cells, including Tm1, 2 and 4. Additional input to T5 comes from Tm9, the medulla target of a lamina interneuron L3. Input synapses from Tm2 and Tm9 are spatially segregated on T5's dendritic arbor, providing candidate anatomical substrates for the two arms of a T5 EMD circuit. Transcript profiling indicates that T5 expresses both nicotinic and muscarinic cholinergic receptors, qualifying T5 to receive cholinergic inputs from Tm9 and Tm2, which both express ChAT. I hypothesize that T5 computes small-field motion signals by integrating multiple cholinergic Tm inputs using nicotinic and muscarinic cholinergic receptors.

T29) Prenatal paternal stress and postnatal enhanced home cage affect maternal care and anxiety like behaviour in juvenile rats

Austin Korgan, Tara Perrot, Ian Weaver

Paternal effects have recently drawn interest from neuroscientists, within the context of epigenetic transmission, largely following evidence from behavioural ecology and epidemiology. Recent studies have demonstrated offspring effects of paternal age, obesity, enrichment, and psychological stress. These outcomes may be partially manifested through differences in maternal care. For the current study, we used predator odour exposure to stress male Long-Evans rats for 7 days, immediately preceding a partner preference test, followed by breeding. On the day of parturition, dams were transferred to enhanced home cages (EHC), which are approximately twice the size of a standard cage (SC) and contain a burrow, or SC. Maternal care was observed for 10 days in both conditions. At P25, two male and two female offspring were observed for social play behaviour for 5 days. Anxiety-like behaviour was examined using the elevated plus maze and open-field test on P32-35. Then, between P35-40 offspring were exposed to a predator odour for 30-minutes and sacrificed for immunohistochemical detection of GH and NPY.

T30) How do student-generated questions relate to comprehension, self-efficacy and achievement in an undergraduate chemistry course?

Bradley Bergey

Since questions can both indicate knowledge deficits and facilitate knowledge construction, it is unclear how student-generated questions relate to comprehension and ability beliefs during classroom learning. This study examines the relations among characteristics of students' questions, perceived comprehension of lectures, self-efficacy beliefs, and exam scores in an undergraduate chemistry course. Participants ($N=103$) recorded questions ($n=980$) in written logs across 8 consecutive lectures. Questions were coded for various qualities. Results from path analyses indicate that different question characteristics have different relations with perceived comprehension and achievement. The percentage of unanswered questions was associated with lower perceived comprehension. Questions that clarified or indicated narrow knowledge deficits were positively associated with perceived comprehension and exam scores. Perceived comprehension predicted situated efficacy beliefs more strongly than did general self-efficacy beliefs for the course. Findings are discussed in light of implications and on-going research.

T31) Dyadic conflict, drinking to cope, and alcohol-related problems: A 4-week longitudinal study

Laura Lambe, Sean MacKinnon, L. Cartwright, M. Korecky, M. Orr, S.J. Park, D. Song, and Sherry Stewart Stewart

The motivational model of alcohol use posits that individuals may consume alcohol to cope with negative affect. Conflict is a strong predictor of coping motives, which consistently predict alcohol-related problems. The current study tested the link between dyadic conflict, coping motives, and alcohol-related problems in romantic couples. It was hypothesized that (a) the association between dyadic conflict and alcohol-related problems would be mediated by coping-depression and coping-anxiety motives, (b) that this would be true for actor and partner effects, and (c) when controlling for enhancement motives. Participants ($N=70$ couples) completed weekly questionnaires for four weeks. Data was analyzed using a longitudinal Actor Partner Interdependence Model. Only coping-depression motives independently mediated the conflict-alcohol problems relationship. This held for actor effects, but not for partner effects. Results remained significant when controlling for enhancement motives. Overall, results suggested certain individuals may be coping maladaptively by turning to alcohol following relationship

T32) Alzheimer's Disease and 5XFAD Mice: A Study of Hormonal Abnormalities and Metabolism

William Gendron, Dr. Richard E. Brown, Dr. Younes Anini

The 5XFAD mouse is a double transgenic model of Alzheimer's disease (AD), which carries an amyloid precursor protein (APP) transgene with three mutations, and a presenilin-1 transgene with two mutations. These mutant transgenes act additively to produce massive increases in beta amyloid (AB)-peptides, and the development of AB-plaques by two months of age. Therefore, the 5XFAD mouse is useful for investigating AB-pathology in AD and for assessing the efficacy of novel therapeutics in reducing AB-plaque load. In our lab, we have previously observed age-related weight-loss in the 5xFAD mouse. Weight-loss is also a common problem in human AD patients. Our objective is to investigate feeding behaviour and feeding-related hormone levels in 5xFAD mice and their WT controls. Several hormones may be involved with this characteristic weight loss, including ghrelin leptin, insulin, thyroid hormones, testosterone, and estrogen. In this study we will collect blood and measure the concentration of the hormones previously stated. Muscle tissue will be collected to evaluate whether mice suffer from sarcopenia (muscle loss), and fat tissue to identify whether weight loss may be attributed with fat loss. We hypothesize that there will be feeding and weight differences, hormonal abnormalities, sarcopenia, all of which indicate an abnormal metabolic state. This project will clarify the mechanisms underlying the feeding abnormalities and weight loss in the 5xFAD mouse model, which parallel human AD symptoms.

T33) The moral bases of John Woolman's opposition to slavery: A comparison of qualitative and quantitative analyses.

John Barresi

How John Woolman came to play a seminal role in the movement to abolish slavery that began in the mid-18th century has recently become an issue of philosophical discussion. My interest is in a psychological account of the moral bases of Woolman's role in this movement. Thus far I have focused on a qualitative analysis of Woolman's moral development using Sylvan Tomkins' *Script Theory* to interpret entries in his *Journal*, an autobiographical document that traces his moral life history and includes discussion of his activities in opposition to slavery. For this talk, I will compare this analysis to findings from a statistical analysis of words used in the *Journal* that correspond to Jonathan Haidt's *Moral Foundations Theory*. The comparison of the two theoretical approaches and empirical methods can provide independent light on the value of each theory as well as a deeper understanding of the origins for Woolman's moral opposition to slavery.

T34) You Can Take a Peek Under the Covers Without Taking a p.

John Christie

Several recent papers have discussed various problems with the way statistics are commonly done including data peeking. It can result in inflation of the Type I error rate to unknown, and sometimes large, amounts. It is strongly discouraged. I discuss how data peeking can be done safely and without any of the typical issues that argue for its abolishment. On the contrary, I argue that proper and disciplined data peeking should be encouraged.

T35) Second-language vocabulary training strategies and their impact on learning outcomes and brain function

Kiera O'Neil, Max Hauser, Vivian Eng, Cheryl Frenck-Mestre and Aaron J Newman

The goal of the proposed study is to assess the efficacy of two second-language (L2) vocabulary training strategies on learning outcomes and brain activation. Words in a language novel to participants (Spanish) will be taught using LANA (Copernicus Studios, Halifax, NS), a set of computer games using automated speech recognition. A vocabulary of 84 words (none of which were English cognates) will be taught over 9, half hour sessions. Half the words will be taught via paired association (i.e. visual representation paired with individual written Spanish word) and half will be taught contextually (i.e. visual representation paired with written Spanish sentences), with word assignment balanced across participants. Participants are to be assessed on Spanish vocabulary knowledge before and after training using both a picture naming task and a picture-word match/mismatch task. ERPs (event-related potentials) will be recorded during the latter task, with the expectation of a larger N400 ERP component post-training, as the N400 is an index of lexical access. After training, subjects should be able to name significantly more items than before training. Mismatched picture-word pairs should elicit an N400 effect after, but not before training. The N400 will be greater for words taught contextually than for those taught by paired association. This research will demonstrate the efficacy of automated speech recognition in teaching vocabulary in a new language, and suggest that contextual teaching strategies appear to foster better second language retention.

T36) Investigating the factors that influence lexical stress processing in English disyllabic words.

Rebecca MacDonald & Erin Sparks

The current study examined adults' processing of letter strings as cues to lexical stress. In English orthography, lexical stress is often associated with patterns of letter strings, typically found in word endings. Adult participants completed a lexical decision task. We selected items that varied in their number of stress friends, i.e. words sharing the same word ending and the same lexical stress. For example, *label* and *novel* are stress friends because they end in *-el* and take first-syllable stress. Using by-items linear regression, we analyzed the impact of the proportion of stress friends, the absolute number of stress friends, and the frequency of stress friends on reaction time. We determined that only the proportion of stress friends predicted reaction time. These findings suggest that adult readers are sensitive to spelling regularities that predict lexical stress and make use of these regularities in their lexical stress processing. We interpret these results within the context of statistical learning theory.

T37) Understanding Cortical Activation Associated with Second Language Acquisition

Thérèse M. Chevalier, Aaron J. Newman

Although transcranial magnetic stimulation (TMS) has been proposed as treatment for individuals with aphasia following a stroke, the function of the brain regions that are the proposed targets of TMS are not well understood. The present study aims to understand cortical activation associated with effortful phonetic processing of a second language in order to add to our understanding of language learning circuitry and provide a basis for further research on treatment of language disorders such as aphasia. Thirty-four healthy adults underwent three computer-based Spanish language-learning sessions, an fMRI scan, and a session of inhibitory TMS, which targeted regions of the left or right hemisphere that were active during fMRI naming tasks. Consistent with our hypothesis, similar patterns of activation were found at the group level during picture naming tasks in English and Spanish. However, no clear patterns emerged following TMS based on stimulation site. Implications for further research will be discussed.

T38) The relationship between implicit sequence learning, speech in noise perception, and working memory.

Alexandria Muise-Hennessey, Sean McWhinney, and Aaron J. Newman.

Language processing involves manipulation and encoding of sequential patterns, and some work has indicated that sequential memory plays an important role in spoken language processing. Indeed, the ability to implicitly learn sequences is correlated with sentence perception in degraded listening conditions (Conway, C. M., Bauernschmidt, A., Huang, S. S., & Pisoni, D. B. (2010). *Cognition*, **114**, 356–71). We investigated whether the relationship between implicit sequential pattern (a Markovian finite-state artificial grammar) learning and speech-in-noise perception (AzBio) is mediated by cognitive functions such as memory, attention, and executive function. Replicating previous work, we found that implicit sequence learning performance was correlated with speech perception in noise. As well, operation span (a working memory measure) was strongly correlated with speech-in-noise perception but not with implicit sequence learning, suggesting that the relationship between sequence learning and speech-in-noise perception is not mediated by working memory. Additional studies are underway to characterize the relationship between speech-in-noise perception and implicit sequence learning as a function of the signal-to-noise ratio of the speech.

T39) Morphological and Syntactic Awareness: Understanding the Deficits in Children with Poor reading Comprehension

S. Helene Deacon, Shelley Tong, & Kate Cain

Poor comprehenders have intact word reading skills but struggle specifically with understanding what they read. We investigated whether two metalinguistic skills, morphological and syntactic awareness, are specifically related to poor reading comprehension. We compared two groups of children in grade 4 matched on multiple skills: poor comprehenders ($n = 15$) and average comprehenders ($n = 15$). The two groups performed comparably on a morphological awareness task that involved both morphological and syntactic cues. However, poor comprehenders performed less well than average comprehenders on a morphological task in which there was no additional syntactic information, and also less well on a syntactic awareness task, in which there were no morphological manipulations. Clearly, poor comprehenders struggle, in comparison to average comprehenders, on tasks targeting each of morphological and syntactic skills separately, but not when the two are combined. Future research needs to identify precisely in what ways these specific metalinguistic difficulties connect to challenges with reading comprehension.

T40) Champ vs. Chump: Knowledge of Opponent Ability Modulates Reward Processing in Medial-Frontal Cortex

Ralph S. Redden, Cameron D. Hassall & Olave E. Krigolson

A series of recent studies (Krigolson & Holroyd, 2006, 2007a, 2007; Krigolson et al., 2008) suggest that a reinforcement learning system within medial-frontal cortex may underlie motor learning. Here, we sought to further that work by examining how expertise modulated reward evaluation by the medial-frontal system. More specifically, we were interested in whether learned knowledge of an opponent's skill level would impact reward processing (i.e. amplitude of the neural response to the presentation of the opponent). In the present study we had participants play rock, paper, scissors against virtual "opponents" while electroencephalographic (EEG) data was recorded. In the key manipulation, participants played three opponents - one who was skilled highly (won 60% of games), skilled poorly (won 20% of games), and skilled moderately (won 33% of games). A post experiment analysis of the EEG data revealed that a reward positivity was elicited when participant wins were contrasted with losses (Holroyd et al., 2008). Furthermore, and importantly, after participants had learned their opponents' ability, viewing "good" opponents faces relative to "bad" opponents faces also yielded a reward positivity. This supports our theory that individuals process opponents differently based on their history of wins and losses against them. These outcomes also support the theory that reward positivity is not only evoked by outcomes, but also modulated by stimuli that predict outcomes, as indicated by a differentiation in medial-frontal activity at the earliest indication of potential reward.

T41) Learning what matters most

Cameron Hassall and Olav Krigolson

Decision making in a complex world is made possible by heuristics, or “mental shortcuts”. One such heuristic, sparsity, guides our choices by focusing learning on only a small subset of available features (such as choosing a graduate program based on how much free food is available). Here, we replicated work suggesting that humans use sparsity when making complex choices, and found electroencephalographic (EEG) evidence implicating selective attention in this process. In particular, we observed an increase in the N2pc (a marker of attention) in response to high-valued relevant features compared to low-valued irrelevant features. Thus, selective attention may be one of the mechanisms behind the use of sparsity in decision making.

T42) The Effect of Visual Noise on the Magnitude of the Directed Forgetting Effect

Noha Mohamed

The present study determined whether the amount of attention paid during initial item processing influenced the strength of the intentional forgetting effect. Using an item-method directed forgetting paradigm, study items in high or low background visual noise appeared one at a time in the centre of a computer screen, each followed by an instruction to remember or forget the previous item. After all study items were presented, participants completed a recognition task in which they indicated whether they recognized the items, regardless of their memory instruction. Results revealed a higher accuracy for to-be-remembered items than to-be-forgotten items. There was no significant difference in the magnitude of the directed forgetting effect between high and low noise conditions. This suggests that the amount of attention paid during encoding is not a significant indicator of the magnitude of the directed forgetting effect. Future research should address both attention withdrawal and cognitive load in the context of directed forgetting effect.

T43) How good is your reaction time data? Lessons learned from the Dalhousie Computerized Attention Battery (DalCAB)

Joshua P. Salmon, Stephanie A. H. Jones, Chris Wright, Michael Lawrence, Beverly Butler, Ray Klein, Gail Eskes

The speed at which a task is performed can be a valuable research instrument in research and clinical assessments. However, standard computer hardware employed for measuring reaction and response times (e.g. a standard computer monitor or mouse) can add noise to the data collection of reaction times, potentially compromising the precision of measurements and subsequent conclusions drawn from the data. Therefore, an assessment of the precision of measurement should be included in the development of computerized tasks and assessments. This talk will outline different methods for assessing the precision of collected reaction time data (with and without independent chronometry). In particular, using example data collected from the Dalhousie Computerized Attention Battery (DalCAB), this talk will discuss the detection, measurement, and correction of non-representative noise in reaction time measurement. This talk should act as a cautionary tale to any researchers who have built their own software, use reaction time as a dependent measure, but have not yet employed methods for verifying the chronometry of their software.

T44) Covert measurement of covert orienting: An electroencephalographic and pupilometric investigation

Mike Lawrence

Covert orienting of spatial attention is typically measured by overt motor responses (keypress, eye movement, etc) made by participants to a peripheral stimulus. This presentation will discuss attempts to simultaneously deploy two measures of covert orienting that do not require such a response, one based on steady-state visual evoked potentials elicited by flickering peripheral stimuli, and another based on a relatively recently discovered effect of covert attention on pupil size when participants view differentially luminous scenes.

T45) Prosodic Factors in Directed Forgetting of Speech Errors

Sarah Dolan, Tracy Taylor, Jonathan Fawcett

Past research has shown that when a speech error is followed by a forget instruction (e.g., “ No, sorry, I meant...”) and then by a correction to the error, participants are able to intentionally forget the error, while performance for the correction to the error is improved as compared to baseline. However, previous research has not controlled for prosodic emphasis of corrections, which is commonly used in natural speech and has been shown to improve memory. We manipulated prosodic emphasis of corrections within auditory narratives containing speech errors. After listening to these narratives, participants were tested for recognition memory. Results showed that recognition performance for errors and control statements was not significantly different, however, performance for corrections was significantly higher than performance for both errors and control statements. We found no impact of prosodic emphasis of corrections on recognition performance. It appears that performance for corrected information benefits from listeners’ increased attention after an error occurs, but that prosodic emphasis does not facilitate this process.

T46) The directed forgetting effect: The processes associated with intentionally forgetting and remembering.

Kate Thompson, Tracy Taylor-Helmick

The processes involved in successfully remembering target information have long been a focus of research in cognitive psychology. Equally important, however, is our ability to intentionally forget irrelevant or misleading information. Investigation of intentional forgetting has progressed through the use of directed forgetting paradigms. In particular, research in the Taylor-Helmick lab has focused on the item-method directed forgetting paradigm. Participants are presented with study items one at a time, each followed by an instruction to Remember or Forget. When tested for memory of ALL items, participants reveal a directed forgetting effect: better memory for items they were told to Remember (R-items) compared to those they were told to Forget (F-items). This difference is the result of a number of cognitive processes -- some that lead to increased memory for R-items, and some that minimize memory for F-items. These processes will be summarized, and the empirical findings that support them will be discussed to provide a complete account of how we remember and forget in item-method directed forgetting.

T47) The Dalhousie Computerized Attention Battery: Validation against traditional neuropsychological testing instruments

Lindsay Rubinfeld

The Dalhousie Computerized Attention Battery is a new battery of experimentally validated tests of attention, developed to measure the vigilance, orienting, and executive control networks of attention as described in a model by Posner and colleagues. The proposed study aims to evaluate the convergent and discriminant validity of the DalCAB with fifty healthy older adult participants (55 years and older) by comparing their performance on the DalCAB to their performance on “gold standard” neuropsychological tests of attention and other cognitive functions using regression analyses. The purpose is to evaluate how well findings from experimentally derived attention measures (i.e., DalCAB tasks) converge with results from clinical measures of attention. Results of this study will further the development of the DalCAB as a tool for the assessment and treatment of attention disorders by clinicians.

T48) Effects of child temperament and parenting on toddlers’ executive functioning

Loriann Williams, Isabel Smith and Nancy Garon

The objective of the current study is to investigate how child temperament and parenting discipline style independently and jointly influence early executive functioning in typically developing preschoolers, observed at two years of age and again, a year later, at three years of age. Specifically, the study addresses: Does a positive parental disciplinary style promote the development of executive function skills in toddlers rated high in temperamental positive affect? It is hypothesized that a) toddlers who are temperamentally positive will display larger gains in executive function abilities at three years of age than toddlers with other temperament profiles and b) that toddlers whose parents engage in positive discipline will exhibit larger executive function gains at three years of age, compared to at two years of age. We also hypothesize that c) parental discipline will mediate the relationship between temperament and executive functioning such that positive discipline during toddlerhood will be related to greater executive functioning for temperamentally positive children at age 3 years. Identifying factors in toddlerhood that may promote subsequent executive function abilities has important implications for social, emotional, and behavioral development. Identification of these factors is also important in the context of community and family-level prevention or early intervention programs aimed at preschoolers.

Special Adjunct Talks

T49) Laterality of Motor Imagery-Related Brain Activity is Modulated by Real-Time Neurofeedback

Tim Bardouille

Motor imagery (MI) is the mental performance of a movement in the absence of actual movement. As in physical practice, the repetitive activation of neural pathways during MI can drive short- and long-term brain changes that underlie skill acquisition. However, the effectiveness of MI may be limited by the lack of feedback from this endogenous process. Real-time feedback about MI-related brain activity (i.e., neurofeedback) has the potential to overcome this limitation. I will present a two-group longitudinal study, wherein participants performed MI across repeated sessions with (group 1) or without (group 2) neurofeedback. Participants who received neurofeedback were able to generate a more contralateral pattern of MI-related brain activity in primary sensorimotor cortex. This indicates that the provision of real-time neurofeedback during MI allows healthy individuals to modulate regional brain activity. This finding has the potential to improve the effectiveness of MI to support skill acquisition and neurorehabilitation.

T50) Integrating methods from neuroscience, experimental psychology, and computer science in the study of cognitive science

Jason Satel

Cognitive science is widely considered to be the last frontier of scientific inquiry. In order to adequately investigate difficult questions in cognitive science, we must begin integrating techniques from multiple disciplines. Although my work so far has focused primarily on the study of visual attention, I am interested in using multiple interdisciplinary methods to investigate questions in cognitive science pertaining to many different topics, including, for example, attention, learning, memory, language, and disorders associated with such functions. In this talk, I will demonstrate how I have integrated methods from neuroscience, psychology, and computer science to investigate a cognitive phenomenon, termed inhibition of return, as an example of how such techniques can be combined to provide insight on difficult topics. I believe this work can be used as a model for the integration of multiple techniques throughout cognitive science and revolutionize the field.

T51) Spelling Matters Too! How spelling practice can benefit reading skill in children
Nicole Conrad

Many theoretical models of reading propose that fluent and skilled reading results from the rapid recognition of letter sequences as a single unit within a word. This rapid recognition depends on previously established memory representations of letter patterns that occur frequently within a language (i.e., orthographic representations). Much research has been devoted to how this orthographic information is acquired incidentally during reading. However, as spelling requires the production rather than recognition of words, spelling may establish higher quality orthographic representation due to the increased thoroughness of processing. During this presentation, I will present studies, in varying stages of completeness, that illustrate gains in spelling can transfer to gains in reading, and that begin to shed light on the mechanisms that enable the acquisition of orthographic knowledge.

POSTERS

Thursday's Posters

P1) fERN differences within Subclinical Psychopaths

Mitchell Coupland & Olav Krigolson

Psychopathy is a personality disorder distinguished by a characteristic lack of affect, superficiality, spontaneous and impulsive behavior, and a parasitic lifestyle. Clinical and subclinical psychopathy differ only in the severity of dysfunction within the individual. One may wonder if psychopathy may impact reward processing within the medial-frontal cortex? The feedback error-related negativity (fERN) is a component of the human event-related brain potential (ERP) that is sensitive to reward processing. In the present study we sought to investigate the relationship between fERN amplitude and subclinical psychopathy; using the Self-Report Psychopathy Scale Revised III to measure levels of subclinical psychopathy within an undergraduate population. Given that, by nature, Psychopaths do not typically learn from their mistakes, we expect to find a deficit in reward processing as reflected in an overall decrease in fERN amplitude for people who score high on psychopathy. Our data supports this contention.

P2) Contrast adaptation in mouse V1 depends on spatial frequency but not firing rate

Matthew P. Lowe, Jillian L. King, Nathan A. Crowder

It is well known that the visual system adjusts its function based on prevailing viewing conditions. For example, prolonged viewing of high contrast gratings produces contrast adaptation in which both the perceived contrast of test stimuli and the responsivity of visual neurons can be altered. Contrast adaptation has been well documented in cats and primates, however its underlying neural mechanisms are not well known. Therefore, we examined this phenomenon in V1 in mice, a genetically manipulable animal model. Previous research in mice has shown that contrast adaptation depends on the spatial frequency (SF) of the adaptor, and that the strength of adaptation can diminish when non-preferred stimuli are used. The purpose of the current study was to determine whether the selectivity of the adaptation effect is due to the SF of the stimuli or merely the sub-maximal spike rates elicited by non-preferred SFs. To do this, we used a top up contrast adaptation protocol where two different adaptors flanked the preferred SF and elicited similar spike rates. The test stimulus either matched the adaptor or was different than the adaptor. Similar to previous studies, we found that contrast adaptation was strongest when the test SF matched the adaptor. It appears that contrast adaptation is mediated by similar underlying neural mechanisms in different animals, which suggests that genetic tools available in mice could be leveraged to investigate these mechanisms.

P3) Retinorecipient areas in the diurnal rodent *Octodon Degu*

Malik Abouleish, Kevin J. Frank, Jillian L. King, Nathan A. Crowder

The degu (*Octodon Degu*) is a diurnal rodent that has been used in neuroscience research to study circadian rhythms. However, degus may also be worth considering as a model in visual neuroscience because they have a far greater density of cone photoreceptors than traditional lab rodents (rats and mice). We used biotinylated dextran amine anterograde labeling to explore the retinofugal projections in the degu for the first time. We documented anterograde labeling in twenty-eight subcortical visual nuclei, but focused the majority of our analysis on the dorsal lateral geniculate nucleus and the superior colliculus as two regions that may have adapted to diurnal activity. We found that the degu's dorsal lateral geniculate nucleus lacked distinct stratification, and was morphologically similar to the rat. The greatest density of labeling we observed occurred in the superior colliculus, with far greater contralateral bias than observed in rats. These differences may reflect the cone-rich retinofugal signal in degus, but overall the morphology of visual projections in this member of the family Octodontidae resembles that in members of the family Muridae.

P4) Investigating a Role for Matrix Metalloproteinases in the Cortex of Visually Deprived Cats.

Josh Bowdridge and Dr. Kevin Duffy

Occlusion of one eye in early life can produce an intractable impairment of visual acuity in the deprived eye that persists through life. Monocular deprivation (MD) in kittens has been studied as a model of human amblyopia, an analogous condition that arises due to abnormal development of the visual system during early postnatal life. Recently, it has been demonstrated that a short period of 10 days of complete darkness is sufficient to reverse the anatomical and functional changes that result from MD. In the present study we chose to examine levels of matrix metalloproteinase (MMP) activity in the visual cortex of animals reared with various forms of abnormal visual experience in an attempt to elucidate the molecular mechanisms that underlie deprivation-induced neural changes, and also those that underlie recovery from MD. Our examination of MMP activity revealed an increase in MMP-2 activity in V1 of kittens after 7 days of monocular deprivation that was coincident with a loss of the MMP substrate, aggrecan.

P5) Recovery from monocular deprivation in kittens induced by ten days of darkness depends upon the acuity in the non-deprived eye and requires a period of binocular vision

Matthew J. Smithen, Kaitlyn Holman, Donald Mitchell & Kevin Duffy

Amblyopia is a condition that results in a deficit in vision and is modelled primarily in kittens by depriving one eye of patterned input for a defined period. Recently, it was discovered that ten days of darkness could promote the recovery of vision in kittens with experimentally induced amblyopia. Experiments were designed, first, to test whether ten days of darkness could promote recovery when the monocular deprivation was extended to eight weeks of age. Second, whether the recovery of vision in the deprived eye was boot-strapped by the neural connections established with the non-deprived eye. Third, if a period of binocular vision was necessary for the recovery seen following ten days of dark rearing.

P6) ERP Components Underlying Prism Adaptation (PA)

Stephane Maclean, Yoko Ishigami-Doyle, Cameron Hassall, Olave Krigolson & Gail Eskes

Prism adaptation (PA) is an experimental technique used to treat spatial neglect after stroke, but the effects are inconsistent over studies, suggesting a need to understand better the underlying mechanisms. Redding and Wallace suggested that the adaptation process involves two components: An immediate strategic recalibration of visuomotor aiming (top-down) and a slower visuo-motor spatial realignment process (bottom-up). The purpose of our study was to examine whether we could detect these components by measuring event-related potentials (ERPs) that are generated during the visuomotor adaptation process induced by exposure to visual field-displacing goggles while reaching. The findings identified two ERPs associated with movement completion that dissociate over the time-course of adaptation: an invariant ERN response sensitive to error commission that could reflect realignment and a P300 response that decreases over time that could reflect conscious visuomotor re-calibration. Implications of these findings for understanding prism adaptation and treatment of neglect will be discussed.

Friday's Posters

P7) Relationship of the components of the Attention Network Test-Interaction (ANT-I) to standardized measures of attention and executive function

Yoko Ishigami, Stewart Longman, Amanda Tyndall, Grazyna Burek, Melanie Denheyer, Marc Poulin, Gail Eskes

The Attention Network Test is a useful computer-based tool for measuring the three attention networks (alerting, orienting, and executive) within a single 20-minute session. We examined relationships between the attention networks and the global performance measured with the Attention Network – Interaction (ANT-I) and standardized neuropsychological tests clinically used to measure different domains of cognitive functioning: attention, processing speed, memory, executive function with healthy older adults (N=181). We found that better executive network scores and better global performance from the ANT-I were associated with processing speed, memory, and executive function as measured by the standardized neuropsychological tests. The results provide insights into the psychometrics of the ANT-I and its potential utility in clinical research settings.

P8) Neuroeconomics: Changes in neural activity during learning

Luke Burke, Laura Alsip, Nicole Hodder, Sarah Vokey, Cameron Hassall, and Olav Krigolson

The purpose of our research was to investigate the brain basis of what it means to be an expert. We recorded EEG while participants completed a perceptual discrimination task and examined two particular components of the event-related potential: the N250 and the reward positivity. Participants showed an enhancement of the N250, thought to index familiarity, for familiar stimuli (previously-learned, computer-generated “blobs”) compared to unfamiliar stimuli (novel blobs). We also observed a decrease in the reward positivity, a positive deflection in response to unexpected rewards, for familiar stimuli compared to unfamiliar stimuli. Thus, in addition to perceptual familiarity, one of the hallmarks of expertise may be that feedback type (correct/incorrect) is more expected, evoking a diminished reward response.

P9) A step towards building better video "games" for rehabilitation: A survey investigating entertainment preferences in adults

Joshua P. Salmon, Graham C. Wilson, Raymond M. Klein, & Gail A. Eskes

As the healthcare system becomes increasingly burdened with demands for rehabilitation services, the need for accessible therapies is growing. Video game approaches offer a promising supplement to existing therapy programs. Research has shown that video games can improve such cognitive functions as memory, processing speed, executive function and attention in both young and older adults. The current challenge however, is finding (or designing) video game approaches that simultaneously match the therapeutic needs of the patient population while remaining fun and engaging. As part of a larger research program, to develop better rehabilitation software or "games" for patients recovering from Stroke or Parkinson's disease, the current study conducted a nation-wide survey of adult entertainment preferences. Measures of favorite television, book, film and movie genres were also collected, as well as information about frequency of use of various technologies, devices and video games. The results from this research offer a window into how adults spend their time with technology, and what approaches in designing new software for older adults might be most effective. The ultimate goal of this research is to apply this information to design more engaging game-like therapies that will encourage and motivate patient rehabilitation.

P10) *Drosophila Melanogaster*

Nedjma Ababou and Meagan Wiederman

Mitochondria produce energy, in the form of ATP, to carry out cellular work. Our experiment hypothesized that larger mitochondrial volumes are present where greater energy demands exist in visual neurons of fruit flies (*Drosophila melanogaster*). We predicted that L1 neurons, the more energetically intensive neurons, would have greater mitochondrial volume and surface area densities than L2 neurons. We predicted that mitochondrial volume would be greatest in the presynaptic area, where neurotransmitter cycling occurs. Using Fiji Image J we traced mitochondria in electron micrograph stacks of four L1 and six L2 neurons to reconstruct the three-dimensional mitochondrial distributions for comparison between neuron types and boutons. Z-tests demonstrated L2 neurons have significantly greater mitochondrial volume and surface area densities; therefore, greater energy requirements. Greater mitochondrial volumes in the postsynaptic terminals of both neurons indicated greater local energy demand. Further research is required to extend this hypothesis to neuron classes in other strata.

P11) The Impact of Socioeconomic Status on Reward Learning in the Medial-Frontal Cortex

Ashley Howse, Marcel Peloquin, S. Doherty, Cam Hassall & Olav Krigolson

While there is strong evidence that a system involved in reward processing exists within the human medial-frontal cortex, there is still a lot to learn about the factors that impact the sensitivity and functional efficacy of this system. Behavioural studies have indicated that children from low socioeconomic (SES) backgrounds may be more sensitive to rewards than those from privileged backgrounds. While children of higher SES outperform their disadvantaged counterparts on cognitive tasks, children of lower SES show greater improvement when rewards are offered for task performance compared to those of higher SES (Sweet & Ringness, 1971; Johnson, 1974). In the present study, we sought to examine the effect of childhood SES on the reward processing system within the human medial frontal cortex. Participants completed a learnable gambling task while electroencephalographic data were recorded. In line with previous work, we found that reward and loss feedback elicited a feedback error-related negativity (fERN; Miltner et al., 1997). Further, and importantly, we found that the amplitude of the fERN is reduced for participants from low SES compared to high SES backgrounds—a result that suggests individuals from low SES backgrounds are more sensitive to rewards than their privileged counterparts. Additionally, a moderate correlation was observed between SES and fERN amplitude suggesting the reward sensitivity decreases as SES increases.

P12) Personality Differences Among Smokers and Non-Smokers

Chad Simmons and Meredith Ivany

Past Research has shown there is a clear link between some personality traits and substance use. Our research will tackle the question of whether there is a measureable difference between the personalities of students that smoke cigarettes and those who do not. We accomplished this by comparing personality dimensions in 432 high school smoking and non-smoking students with the Substance Use Risk Profile Scale (SURPS) and the Connor-Davidson Resilience Scale 10 (CD-RISC). The SURPS measured hopelessness (HI), anxiousness (AS), impulsivity (IMP) and sensation seeking (SS). The CD-RISC measured resiliency. Smokers were found to have higher hopelessness and sensation seeking as well as lower resiliency scores than non-smokers, and within the smokers, those with high resiliency started smoking later in life than those with low resiliency. Smokers with high HI attended fewer smoking cessation sessions, and those who completed the program were found to have lower HI and SS scores. This suggests that individuals with different personality traits would benefit from different treatment types, for example, smokers with high HI may benefit from a focus on self-worth in their treatment. Furthermore, high resiliency appears to delay or prevent the onset of smoking, so increasing resiliency in our children is a step that can be taken in the future.